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IN-WHEEL MOTOR SYSTEM

Abstract

An in-wheel motor system includes a fixed part, a rotating part configured to be rotatable relative to the fixed part, and a brake mounted on the fixed part. The brake includes a friction material disposed to contact the rotating part.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims, under 35 U.S.C. § 119(a), the benefit of priority to Korean Patent Application No. 10-2024-0031163, filed Mar. 5, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an in-wheel motor system.

BACKGROUND

[0003] Electric vehicles driven by motors instead of engines have received considerable attention in recent years. Generally, electric vehicles are configured to be driven by one or two large motors. In-wheel motors widely used in small vehicles such as electric bicycles and scooters are being applied to electric cars.

[0004] An in-wheel motor is an electric motor mounted in a wheel of a vehicle and configured to directly rotate the wheel. Specifically, in an electric vehicle driven by the in-wheel motor, a small motor is mounted in each wheel to independently drive and control each wheel.

[0005] The electric vehicle having the in-wheel motor mounted therein may provide various advantages such as excellent space utilization and improved control performance compared to an electric vehicle having an existing large motor mounted therein. However, since the electric vehicle driven by the in-wheel motor has a motor provided in each wheel, it is necessary to meet several design conditions different from design conditions of an electric vehicle driven by one large motor.

[0006] The above information disclosed in this Background section is only for enhancement of understanding of the background of the disclosure, and therefore it may contain information that does not form the prior art that is already known to a person of ordinary skill in the art.

SUMMARY

[0007] The present disclosure has been made in an effort to solve the above-described problems associated with the prior art. It is an object of the present disclosure to provide an in-wheel motor system configured to reduce a size of an in-wheel motor and also to enable high output.

[0008] It is another object of the present disclosure to provide an in-wheel motor system having improved efficiency.

[0009] The objects of the present disclosure are not limited to the above-mentioned objects. Other technical objects not mentioned herein should be more clearly understood by one having ordinary skill in the art to which the present disclosure pertains from the detailed description of the embodiments.

[0010] In one aspect, the present disclosure provides an in-wheel motor system including a fixed part, a rotating part configured to be rotatable relative to the fixed part, and a brake mounted on the fixed part. The brake includes a friction material disposed to contact the rotating part.

[0011] Other aspects and embodiments of the present disclosure are discussed below.

[0012] It should be understood that the terms “vehicle”, “vehicular”, and other similar terms as used herein are inclusive of motor vehicles in general. Such motor vehicles may encompass passenger automobiles including sport utility vehicles (SUV), buses, trucks, various commercial vehicles, watercraft including a variety of boats and ships, aircraft, and the like. Such motor vehicles may also include hybrid vehicles, electric vehicles, plug-in hybrid electric vehicles, hydrogen-powered vehicles, and other alternative fuel vehicles (e.g., fuels derived from resources

other than petroleum). As referred to herein, a hybrid vehicle is a vehicle that has two or more sources of power, for example, vehicles powered by both gasoline and electricity.
[0013] The above and other features of the disclosure are discussed below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The above and other features of the present disclosure are described in detail with reference to embodiments thereof illustrated in the accompanying drawings which are given hereinbelow by way of illustration only, and thus are not limitative of the present disclosure.

[0015] FIG. 1 is a cross-sectional view of an in-wheel motor system according to an embodiment of the present disclosure.

[0016] FIG. 2 is a cross-sectional view of a portion of the in-wheel motor system according to an embodiment of the present disclosure.

[0017] FIG. 3 is a perspective view of a main body of a modular coil for the in-wheel motor system according to an embodiment of the present disclosure.

[0018] FIG. 4 is a perspective view of the modular coil for the in-wheel motor system according to an embodiment of the present disclosure.

[0019] FIG. 5 is a view showing a stator of the in-wheel motor system including the modular coil according to an embodiment of the present disclosure.

[0020] FIG. 6 is a view showing a protective body for the modular coil according to an embodiment of the present disclosure.

[0021] FIG. 7 is an unfolded view of FIG. 6.

[0022] FIG. 8 is an exploded perspective view of the modular coil and the protective body according to an embodiment of the present disclosure.

[0023] FIG. 9 is a view showing a process of mounting the modular coil and the protection body according to an embodiment of the present disclosure on the stator of the in-wheel motor system.

[0024] FIG. 10 is a view showing some components of the in-wheel motor system including the modular coil according to an embodiment of the present disclosure.

[0025] FIG. 11 is a cross-sectional view when the components in FIG. 10 are combined.

[0026] FIG. 12 is an enlarged view of a portion S1 in FIG. 11.

[0027] FIG. 13 is an enlarged view of a portion S2 in FIG. 11.

[0028] FIG. 14 is a view showing a cooling housing of the in-wheel motor system according to an embodiment of the present disclosure.

[0029] FIG. 15 is a view showing a cooling channel formed in the cooling housing of the in-wheel motor system according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0030] Specific structural or functional descriptions given in connection with the embodiments of the present disclosure are merely illustrative for the purpose of describing embodiments according to the concept of the present disclosure. The embodiments according to the concept of the present disclosure may be implemented in various forms. Further, it should be understood that the present description is not intended to limit the disclosure to the embodiments. On the contrary, the disclosure is intended to cover not only the embodiments, but also various alternatives, modifications, equivalents, and other embodiments, which may be included within the spirit and scope of the disclosure as defined by the appended claims.

[0031] In the present disclosure, terms such as “first” and/or “second” may be used to describe various components, but the components are not limited by the terms. The terms are used only for the purpose of distinguishing one component from other components. For example, a first component may be referred to as a second component, and similarly, a second component may also

be referred to as a first component without departing from the scope of rights according to the concept of the present disclosure.

[0032] When one component is referred to as being “connected” or “joined” to another component, the one component may be directly connected or joined to the other component, but it should be understood that other components may be present therebetween. On the other hand, when the one component is referred to as being “directly connected to” or “directly in contact with” the other component, it should be understood that other components are not present therebetween. Other expressions for the description of relationships between components, i.e., “between” and “directly between” or “adjacent to” and “directly adjacent to,” should be interpreted in the same manner.

[0033] The same reference numerals represent the same components throughout the specification. Additionally, the terms in the specification are used merely to describe embodiments and are not intended to limit the present disclosure. In this specification, an expression in a singular form also includes a plural form, unless clearly specified otherwise in context. As used herein, expressions such as “comprise” and/or “comprising” do not exclude the presence or addition of one or more components, steps, operations, and/or elements other than those described. The same understanding should apply to terms such as “have”, “include”, and the like, and variations thereof.

[0034] Hereinafter, the present disclosure is described in detail with reference to the accompanying drawings. When a component, device, element, or the like of the present disclosure is described as having a purpose or performing an operation, function, or the like, the component, device, or element should be considered herein as being “configured to” meet that purpose or perform that operation or function.

[0035] As shown in FIG. 1, an in-wheel motor system **1** according to an embodiment of the present disclosure may be mounted in a wheel **10** of a vehicle. The x-axis direction refers to the lateral direction of the vehicle, the y-axis direction refers to the vertical direction, and the z-axis direction refers to the longitudinal direction of the vehicle. Accordingly, the in-wheel motor system **1** in FIG. 1 is installed on the wheel on the driver's seat side of the vehicle. In other words, the left side of FIG. 1 is the outer side of the vehicle, and the right side is the inner side of the vehicle.

[0036] The in-wheel motor system **1** is configured to be rotatable within the wheel **10** to directly rotate the wheel **10**. To this end, the in-wheel motor **1** includes a rotating part and a fixed part.

[0037] The rotating part is configured to be rotatable relative to the fixed part. The rotating part is configured to be rotatable, particularly, by electromagnetic interaction between a rotor **20** serving as the rotating part and a stator **30** serving as the fixed part. In one embodiment, the in-wheel motor system **1** may be an outer rotor type motor in which the rotor **20** is disposed radially outward relative to the stator **30**.

[0038] The rotor **20** includes a rotor core **22** and a first magnetic member **24**. One or more first magnetic members **24** may be mounted on the rotor core **22**. The stator **30** includes a stator core **32** and a second magnetic member **34**. One or more second magnetic members **34** may be mounted on the stator core **32**.

[0039] In some embodiments, both the first magnetic member **24** and the second magnetic member **34** may be electromagnets configured to allow current to be applied thereto. In some embodiments, any one of the first magnetic member **24** and the second magnetic member **34** may be a permanent magnet, and the other one may be an electromagnet. For example, the first magnetic member **24** may be a permanent magnet, and the second magnetic member **34** may be an electromagnet. In this specification, a description is given as to an example in which the first magnetic member **24** of the rotor **20** is a permanent magnet, and the second magnetic member **34** of the stator **30** is an electromagnet.

[0040] The first magnetic member **24** and the second magnetic member **34** may interact electromagnetically by magnetization of the second magnetic member **34** of the stator **30**. When a current supplied to the second magnetic member **34** is adjusted, the rotor **20** including the first magnetic member **24** may be rotatable around a centerline C of the wheel **10** relative to the stator

30 including the second magnetic member **34**.

[0041] The rotating part may further include an inner cover **40** and a shaft **50**. The shaft **50** may be coupled to the center of the rotor **20**, and the shaft **50** may be coupled to the inner cover **40**. The inner cover **40** may prevent foreign substances from flowing into the in-wheel motor system **1** and may protect components, such as a brake **80** and a wheel bearing **60**. Which are positioned radially inside the stator **30**. Additionally, the wheel **10** having a tire **12** mounted thereon may be coupled to the shaft **50**. For example, the rotor **20**, the inner cover **40**, the shaft **50**, and the wheel **10** may be coupled to each other by a coupling member (b), which may be detachably coupled to each other. In some embodiments, the rotor **20**, the inner cover **40**, and the shaft **50** may be integrally formed with each other without the coupling member (b).

[0042] The rotating part is rotatably coupled to the fixed part. Specifically, the shaft **50** of the rotating part is rotatably disposed relative to the fixed part by the wheel bearing **60**. In addition to the stator **30**, the fixed part may include a support bracket **70**, the wheel bearing **60**, the brake **80**, and a cooling housing **160**.

[0043] The stator **30** may be fixedly connected to a vehicle by the support bracket **70**. For example, the support bracket **70** may be detachably coupled to the stator core **32** through a coupling member (not shown). Also, the wheel bearing **60** and the brake **80** may be detachably coupled to the support bracket **70**. For example, the coupling member (b) may detachably couple the support bracket **70**, the wheel bearing **60**, and the brake **80** to each other.

[0044] The in-wheel motor system **1** further includes an annular seal **90**. The seal **90** may include an outer seal **92** and an inner seal **94**. A space between the rotor **20** and the stator **30** is sealed by the seal **90** on opposite sides in the x-axis direction or the axial direction of the rotor **20** and the stator **30**. In one embodiment, the seal **90** includes the outer seal **92** disposed on the outer side of a vehicle in the lateral direction x and the inner seal **94** located closer to the center of the vehicle than the outer seal **92**. As a non-limiting example, the seal **90** may be made of rubber.

[0045] According to an embodiment of the present disclosure, the in-wheel motor system **1** includes the brake **80**. The brake **80** may be coupled to the radially outer side of the wheel bearing **60**. According to an embodiment of the present disclosure, the brake **80** may include a cylinder **82** and a friction material **84**. The cylinder **82** may be a hydraulic cylinder.

[0046] Referring to FIG. 2, the brake **80** may operate in conjunction with the inner cover **40**. The friction material **84** of the brake **80** may use an inner surface **42** of the inner cover **40** as a friction surface. Through this configuration, the brake **80** may be provided in the in-wheel motor system **1** without an additional increase in size.

[0047] When the cylinder **82** is operated in response to an operation request of the brake **80**, the friction material **84** contacts the inner surface of the inner cover **40**, thereby generating braking force. Also, the brake **80** may further include a return spring (not shown). The return spring may provide force to return the friction material **84** to the original position thereof when the operation of the brake **80** is terminated. The in-wheel motor system **1** according to an embodiment of the present disclosure utilizes the inner cover **40** and unused space in an in-wheel motor, thereby making it possible not only to prevent an increase in the size of the in-wheel motor system **1**, but also to secure effects, such as sound insulation and rust prevention, through the inner cover **40**. Through the above-described structure, the in-wheel motor system **1** according to an embodiment of the present disclosure may be compactly designed, and an in-wheel motor may be applied to wheels of 16 inches or less. In the related art, in order to implement an in-wheel motor configured to generate high output, it is required to apply an in-wheel motor only to wheels of 19 inches or larger due to an inevitable increase in size.

[0048] Referring to FIG. 3, the in-wheel motor system **1** may include a modular coil **100**. The modular coil **100** may be applied as the second magnetic member **34**.

[0049] In the modular coil **100**, a coil **101** is wound to form a main body **103** based on a preset shape. As used herein, the main body **103** may mean the coil **101** formed in the preset shape. In

addition, the coil **101** may be a flat plate coil. The flat plate coil **101** may increase magnetic flux density by minimizing an empty space and improving an occupying ratio of the coil. In some embodiments, the main body **103** may be formed by winding the coil **101**. In some embodiments, the main body **103** may be formed by three-dimensionally printing a coil material.

[0050] In an embodiment shown in the drawing, an end **1101** of the coil **101** is configured to protrude outward from the main body **103** for an electrical connection with other components. A first end **1101a** and a second end **1101b** of the coil **101** may be formed on the same side of the main body **103** and may be extended in the same direction in a state of being spaced apart from each other.

[0051] As shown in FIG. 4, the main body **103** includes a coating layer **105**. The coating layer **105** may mean a molding layer or a coating layer. The coating layer **105** is configured to insulate the main body **103** and protect the main body **103**. In addition, the moldable coating layer **105** may surround the main body **103** in a shape that allows easy assemble of the modular coil **100**.

According to an embodiment of the present disclosure, the coating layer **105** may prevent an electrical short that may occur when the coils **101** are stacked and may improve durability of the coil **101**.

[0052] After the coating layer **105** surrounds the main body **103**, the first end **1101a** and the second end **1101b** are adapted to be connected to a bus bar **150** for electrical connection. A part of the first end **1101a** and the second end **1101b** each extending from the main body **103** may be configured not to be surrounded by the coating layer **105**.

[0053] As shown in FIG. 5, the modular coil **100** may be mounted on the stator core **32** of the stator **30**. The modular coil **100** may be mounted on a corresponding one of teeth **36** provided in the stator core **32**. The teeth **36** may be formed to be spaced apart from each other by a predetermined gap along the circumference of the stator core **32**. In some embodiments, when the in-wheel motor system **1** is an outer rotor type motor, the teeth **36** are configured to extend outward in the radial direction of the stator core **32** from the stator core **32**.

[0054] The modular coil **100** may be mounted on a corresponding one of the teeth **36** through a protective body. The protective body may improve insulation performance of the modular coil **100**. In some embodiments, the protective body may include a protective sheet **107**. As a non-limiting example, the protective sheet **107** may be insulating paper. Each protective sheet **107** is molded in a shape capable of surrounding a corresponding one of the teeth **36**.

[0055] Referring to FIG. 6, in another embodiment, the protective body may include the protective sheet **107** and a support **109**. Each protective sheet **107** is molded into a predetermined shape, capable of surrounding a corresponding one of the teeth **36**, and the support **109** is attached to opposite sides of the protective sheets **107**. For example, the support **109** may be coupled to the protective sheets **107** by an adhesive.

[0056] As shown in FIG. 7, folding lines **L1** and **L2** are arranged in the protective sheet **107**. The folding lines **L1** and **L2** include the longitudinal folding line **L1** extending in the longitudinal direction of the protective sheet **107** and the lateral folding line **L2** extending perpendicular to the longitudinal folding line **L1**.

[0057] In an embodiment, a seating part **1107** is provided between the lateral folding lines **L2** of the protective sheet **107**. The support **109** is configured to be attached to the seating part **1107**. In one embodiment, grooves **2107** may be respectively provided at opposite ends of the seating part **1107**. Each of the grooves **2107** is provided to indicate the position of the support **109** so that an attachment position of the support **109** is easily determined.

[0058] As shown in FIG. 8, after the support **109** is coupled to the protective sheet **107**, the protective sheet **107** is formed along the lateral folding line **L2**. The protective sheet **107** folded in this manner may be inserted into the modular coil **100**.

[0059] Referring to FIG. 9, the inserted protective sheet **107** is folded toward the modular coil **100** along the longitudinal folding line **L1**. Accordingly, the modular coil **100** may contact the stator

core **32** through the protective sheet **107**. When the protective sheet **107** is completely folded, the modular coil **100** may be inserted into a corresponding one of the teeth **36**.

[0060] As shown in FIGS. **10** and **11**, according to an embodiment of the present disclosure, the modular coil **100** mounted on the stator core **32** may be fixed and protected by the components of the in-wheel motor system **1**. A front cover **110** is disposed at one side of the stator **30**. The front cover **110** is disposed to cover a first side of the modular coil **100**. The front cover **110** may be fixed by a seal housing **120** for the outer seal **92**. The outer seal **92** is disposed in the seal housing **120** so as to prevent foreign substances from the outside from flowing into the motor. The seal housing **120** may be coupled to the cooling housing **160** disposed radially inside the stator core **32** through a bolt **130**. Referring to FIG. **12**, in some embodiments, the seal housing **120** may have a coupling part **122** provided therein and the front cover **110** may have an accommodation part **112** provided therein. When the coupling part **122** and the accommodation part **112** are engaged with each other, the front cover **110** may be fixed.

[0061] Referring to FIG. **13**, a rear cover **140** is disposed at a second side of the stator **30**. The second side of the stator **30** may be an opposite end of the first side. The rear cover **140** is disposed to cover the second side of the modular coil **100**. A bus bar **150** for electrical connection is mounted on the second side of the modular coil **100**. The rear cover **140** may accommodate the bus bar **150** and the modular coil **100** therein. In this manner, the bus bar **150** and the modular coil **100** are located in the rear cover **140**. The rear cover **140** may be fixed by the cooling housing **160**. In some embodiments, the rear cover **140** may be a cover for a busbar.

[0062] Referring to FIG. **14**, according to an embodiment of the present disclosure, the in-wheel motor system **1** may include the cooling housing **160**. The cooling housing **160** is disposed radially inside the stator **30**. Additionally, the cooling housing **160** is disposed adjacent to the inner cover **40**. In addition to cooling the stator **30**, the cooling housing **160** may cool heat generated in the inner cover **40** by operation of the brake **80**. In some embodiments, the cooling housing **160** may be integrated with the stator core **32**. Alternatively, the cooling housing **160** and the stator core **32** may be separately formed and disposed to contact each other. Hereinafter, a description is given as to an example in which the cooling housing **160** is integrated with the stator core **32**.

[0063] The cooling housing **160** includes a cooling channel **162**. The cooling channel **162**, which is a passage through which a coolant may flow, is formed in the stator core **32** or the cooling housing **160**. In this example, the coolant flows through an inlet **164** and an outlet **166**. The coolant flows through the inlet **164** where the coolant flows in and the outlet where the coolant after cooling is discharged.

[0064] In order to improve cooling performance, a coolant capacity or an area may be increased, or a specially designed water jacket structure having high cooling performance may be applied. In the former case, it is required to increase a thickness of the cooling channel. In this case, an increase in part size is required, which may lead to an insufficient space problem. Further, in this case, a space allocated to each part is reduced, which may cause deterioration in motor performance. In the latter case, cooling performance is improved, but it is difficult to manufacture the specially designed water jacket structure.

[0065] When the temperature of an internal winding is not uniform, output and durability of the in-wheel motor may be affected. Therefore, it is necessary to uniformly cool the inside of the in-wheel motor system **1**. According to an embodiment of the present disclosure, the above-described problems may be solved by applying a U-turn flow route to the cooling channel **162**, as described below.

[0066] Referring to FIG. **15**, the cooling channel **162** is continuously formed as one channel. In other words, after the coolant introduced through the inlet **164** passes through the entire cooling channel **162**, the coolant is discharged through the outlet **166** to form the U-turn flow route.

[0067] The cooling channel **162** has flow paths in various directions. The cooling channel **162** may extend in the circumferential direction of the stator core **32** and may extend in the axial direction of

the stator core **32**. The cooling channel **162** may change a direction thereof in the circumferential direction and the axial direction of the stator core **32** and may extend within the stator core **32**, thereby providing a coolant flow path formed in various directions. The cooling channel **162** having the above-mentioned shape allows the coolant to flow or move in various directions at various locations within the stator core **32**. To this end, the cooling channel **162** may include a plurality of channel portions **1162**, **2162**, **3162**, **4162**, **5162**, **6162**, and **7162**.

[0068] According to an embodiment of the present disclosure, the cooling channel **162** may include the outer channel portions **1162** and **2162** and the inner channel portions **3162** and **4162**. The outer channel portions **1162** and **2162** and the inner channel portions **3162** and **4162** are continuously connected to each other.

[0069] The outer channel portions **1162** and **2162** include the first outer channel portion **1162** extending in the circumferential direction of the stator core **32** from a side at the circumferential side of the stator core **32**, i.e., from a first side of the stator core **32**. Additionally, the outer channel portions **1162** and **2162** include the second outer channel portion **2162** extending in the circumferential direction of the stator core **32** from a second side opposite the first side of the stator core **32**. The outer channel portions **1162** and **2162** may be disposed close to the outer periphery of the stator core **32**.

[0070] The first outer channel portion **1162** and the second outer channel portion **2162** are connected to each other by the first connection portion **5162**. The first outer channel portion **1162** and the second outer channel portion **2162** may extend in the circumferential direction of the stator core **32**. The first connection portion **5162** may extend in the axial direction of the stator core **32**. Accordingly, the first outer channel portion **1162** and the second outer channel portion **2162** may be spaced apart from each other by the first connection portion **5162**. The flow direction of the coolant may be reversed in the channel portions **1162** and **2162**.

[0071] The second outer channel portion **2162** is connected to the inner channel portions **3162** and **4162** by the second connection portion **6162**. The inner channel portions **3162** and **4162** are disposed between the outer channel portions **1162** and **2162**. In this manner, the first inner channel portion **3162** and the second inner channel portion **4162** may be provided.

[0072] The first inner channel portion **3162** is connected to the second outer channel portion **2162** by the second connection portion **6162**. The first inner channel portion **3162** extends in the circumferential direction of the stator core **32**. The second connection portion **6162** extends in the axial direction of the stator core **32**. Accordingly, the flow direction of the coolant in the cooling channel **162** may be changed through the second connection portion **6162**. The first inner channel portion **3162** may allow the coolant to flow in a direction opposite the second outer channel portion **2162**.

[0073] The first inner channel portion **3162** is connected to the second inner channel portion **4162**. The third connection portion **7162** is provided between the first inner channel portion **3162** and the second inner channel portion **4162** to change the flow direction of the coolant. The third connection portion **7162** extends in the axial direction of the stator **30** from the second inner channel portion **4162**. The second inner channel portion **4162** connected to the third connection portion **7162** extends in the circumferential direction of the stator core **32**.

[0074] In other words, if the coolant flows in a first direction in the first outer channel portion **1162**, the coolant flows in a second direction opposite the first direction in the second outer channel portion **2162**. The coolant is configured to flow again in the first direction in the first inner channel portion **3162** and to flow again in the second direction in the second inner channel portion **4162**. The flow direction of the coolant may be changed through the connection portions **5162**, **6162**, and **7162** each having a curved portion in the cooling channel **162**.

[0075] In the drawing, a portion connected to the second inner channel portion **4162** is indicated as the inlet **164** into which the coolant is introduced, and a portion connected to the first outer channel portion **1162** is indicated as the outlet **166** through which the coolant is discharged. However, the

present disclosure is not limited thereto. The first outer channel portion **1162** may be connected to the inlet **164** and the second inner channel portion **4162** may be connected to the outlet **166**.
[0076] The cooling channel **162** may perform uniform cooling at a high heating temperature of a coil, such as 200 degrees Celsius or higher.
[0077] The in-wheel motor system **1** according to an embodiment of the present disclosure may enable high output while reducing the size of the in-wheel motor through the modular coil **100**, the brake **80**, and the cooling channel **162** described above.
[0078] As is apparent from the above description, the present disclosure provides an in-wheel motor system configured not only to reduce a size of an in-wheel motor but also to enable high output.
[0079] Additionally, the present disclosure provides an in-wheel motor system having improved efficiency.
[0080] The effects of the present disclosure are not limited to the above-mentioned effects. Other effects not mentioned herein should be more clearly understood by those having ordinary skill in the art from the detailed description of the embodiments.
[0081] The present disclosure described above is not limited to the above-described embodiments and the accompanying drawings. However, it should be appreciated by those having ordinary skill in the art that changes may be made in these embodiments without departing from the principles and spirit of the disclosure, the scope of which is defined in the appended claims and equivalents thereto.

Claims

1. An in-wheel motor system comprising: a fixed part; a rotating part configured to be rotatable relative to the fixed part; and a brake mounted on the fixed part, the brake comprising a friction material disposed to contact the rotating part.
2. The in-wheel motor system of claim 1, wherein the rotating part comprises: a rotor comprising a permanent magnet; and an inner cover disposed inside the rotor and configured to contact the friction material.
3. The in-wheel motor system of claim 2, wherein the friction material is configured to contact an inner surface of the inner cover.
4. The in-wheel motor system of claim 2, wherein the brake further comprises a cylinder configured to move the friction material to enable the friction material to contact the inner cover.
5. The in-wheel motor system of claim 2, wherein the fixed part comprises a stator disposed inside the rotor, wherein a coil is wound around the stator.
6. The in-wheel motor system of claim 1, wherein the fixed part comprises a wheel bearing configured to rotatably support the rotating part and wherein the brake is mounted on the wheel bearing.
7. The in-wheel motor system of claim 1, further comprising a seal disposed between the fixed part and the rotating part and configured to seal a space between the fixed part and the rotating part.
8. The in-wheel motor system of claim 1, wherein the fixed part comprises a stator having a modular coil mounted thereon and wherein a current is applicable to the modular coil.
9. The in-wheel motor system of claim 8, wherein the modular coil comprises: a coil main body having a flat plate coil formed in a preset shape; and a coating layer configured to surround the coil main body.
10. The in-wheel motor system of claim 8, wherein the stator comprises teeth spaced apart from each other along a circumference of the stator and wherein the modular coil is mounted on each of the teeth.
11. The in-wheel motor system of claim 8, further comprising an insulating paper disposed between the modular coil and the stator.

- 12.** The in-wheel motor system of claim 8, wherein the stator comprises a cooling housing comprising a cooling channel configured to allow a coolant to flow through the stator.
- 13.** The in-wheel motor system of claim 12, wherein: the cooling channel comprises an inlet and an outlet, wherein the inlet allows the coolant to flow thereinto, and the outlet allows the coolant to be discharged therethrough; and the cooling channel is continuous from the inlet to the outlet.
- 14.** The in-wheel motor system of claim 13, wherein the cooling channel is a single channel from the inlet to the outlet.
- 15.** The in-wheel motor system of claim 12, wherein the cooling channel extends in a circumferential direction of the stator and an axial direction of the stator.
- 16.** The in-wheel motor system of claim 12, wherein the cooling housing is integrally formed with the stator.
- 17.** The in-wheel motor system of claim 12, wherein the cooling housing is disposed between the modular coil and the brake.
- 18.** A vehicle driven by the in-wheel motor system of claim 1.
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